```python
import pandas as pd

df = pd.read_csv("all_stocks_5yr.csv")

print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
```

```
Dataset Shape: (619040, 7)

First 5 rows:
         date   open   high    low  close    volume Name
0  2013-02-08  15.07  15.12  14.63  14.75   8407500  AAL
1  2013-02-11  14.89  15.01  14.26  14.46   8882000  AAL
2  2013-02-12  14.45  14.51  14.10  14.27   8126000  AAL
3  2013-02-13  14.30  14.94  14.25  14.66  10259500  AAL
4  2013-02-14  14.94  14.96  13.16  13.99  31879900  AAL
```

```python
stock = df[df["Name"] == "AAL"].copy()

stock["date"] = pd.to_datetime(stock["date"])


stock = stock.sort_values("date")


print("\nMissing Values:")
print(stock.isnull().sum())


print("\nAAL Stock Data:")
print(stock.head(10))

print("\nNumber of AAL records:", len(stock))
```

```
Missing Values:
date      0
open      0
high      0
low       0
close     0
volume    0
Name      0
dtype: int64

AAL Stock Data:
        date   open   high    low  close    volume Name
```

```
0 2013-02-08  15.07  15.12  14.63  14.75   8407500  AAL
1 2013-02-11  14.89  15.01  14.26  14.46   8882000  AAL
2 2013-02-12  14.45  14.51  14.10  14.27   8126000  AAL
3 2013-02-13  14.30  14.94  14.25  14.66  10259500  AAL
4 2013-02-14  14.94  14.96  13.16  13.99  31879900  AAL
5 2013-02-15  13.93  14.61  13.93  14.50  15628000  AAL
6 2013-02-19  14.33  14.56  14.08  14.26  11354400  AAL
7 2013-02-20  14.17  14.26  13.15  13.33  14725200  AAL
8 2013-02-21  13.62  13.95  12.90  13.37  11922100  AAL
9 2013-02-22  13.57  13.60  13.21  13.57   6071400  AAL

Number of AAL records: 1259
```

```
stock["prev_close"] = stock["close"].shift(1)

stock["MA5"] = stock["close"].rolling(5).mean()

stock["target"] = stock["close"].shift(-1)

stock = stock.dropna()

X = stock[["open", "high", "low", "close", "volume", "prev_close",
"MA5"]]

y = stock["target"]

print("Features:")
print(X.head())

print("\nTarget:")
print(y.head())

print("\nFeature Shape:", X.shape)
print("Target Shape:", y.shape)
```

```
Features:
    open   high    low  close    volume  prev_close     MA5
4  14.94  14.96  13.16  13.99  31879900       14.66  14.426
5  13.93  14.61  13.93  14.50  15628000       13.99  14.376
6  14.33  14.56  14.08  14.26  11354400       14.50  14.336
7  14.17  14.26  13.15  13.33  14725200       14.26  14.148
8  13.62  13.95  12.90  13.37  11922100       13.33  13.890

Target:
4    14.50
5    14.26
6    13.33
7    13.37
8    13.57
```

```
Name: target, dtype: float64

Feature Shape: (1254, 7)
Target Shape: (1254,)



split = int(len(stock) * 0.8)

X_train = X.iloc[:split]
X_test = X.iloc[split:]

y_train = y.iloc[:split]
y_test = y.iloc[split:]

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)

Training data: (1003, 7)
Testing data: (251, 7)

from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, y_train)

print("Model training completed!")

Model training completed!


y_pred = model.predict(X_test)

print("Predicted Stock Prices:")
print(y_pred[:10])

Predicted Stock Prices:
[45.01231229 45.53054436 46.31971686 46.79236993 46.88260057
47.24228482
 47.21255522 46.88552264 46.92234906 46.51616933]

from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
import numpy as np

mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("Model Evaluation")
print("-----------------")
```

```
print("MAE :", mae)
print("RMSE:", rmse)
print("R²  :", r2)
```

```
Model Evaluation
-----------------
MAE : 0.7869399560159042
RMSE: 1.0403287010433837
R²  : 0.9254466096187646
```

```
result = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print(result.head(10))
```

```
   Actual  Predicted
0   46.30  45.012312
1   46.45  45.530544
2   47.41  46.319717
3   46.57  46.792370
4   47.54  46.882601
5   46.97  47.242285
6   46.91  47.212555
7   46.81  46.885523
8   46.32  46.922349
9   46.10  46.516169
```

```
latest_data = X.iloc[[-1]]

next_day_prediction = model.predict(latest_data)

print("Predicted Next Day Closing Price:", next_day_prediction[0])
```

```
Predicted Next Day Closing Price: 50.4409970868843
```